

NATIONAL UNIVERSITY OF SINGAPORE

EE2213 - Introduction to Artificial Intelligence

Semester 1: AY 2025/26

Quiz 3: L7-L11

Time Allowed: 30 minutes

INSTRUCTIONS TO STUDENTS

1. This assessment consists of **FOUR** types of questions: True/False Questions, Multiple Choice Questions (MCQs, only one correct answer), Multiple Response Questions (MRQs, allow multiple correct answers), and Fill-in-the-Blank-Questions (FITB).
2. There are a total of **12 Questions** consisting of **2** True/False Questions (*2 marks each*), **2** Multiple Choice Questions (*3 marks each*), **5** Multiple Response Questions (*5 marks each*), and **3** Fill-in-the-Blank Questions (*7 marks each*) for a total of **56** marks.
3. Please answer **ALL** questions.
4. This is a **NON-SECURE BLOCK INTERNET** assessment. You may use printed or soft copy materials (e.g., lecture slides, textbooks, notes). However, you are not allowed to search the internet or use any LLM-powered tools (e.g., ChatGPT or other language models)
5. Electronic devices other than your PC should be left on the floor next to your desk or put inside your bags at all times. Please note that graphing calculators are not permitted. A basic calculator is provided for numerical calculations, accessible under “Tool Kit” at the top right of the exam interface
6. You are not permitted to leave the venue for the first 15 minutes.
7. If you wish to communicate with an invigilator, go to the washroom, or leave before the end of the assessment, please raise your hand to inform the invigilator.
8. You will be liable for disciplinary action which may result in expulsion from the University if you are found to have contravened any of the clauses below,
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- Bringing your mobile phone or any storage/communication device with you to the washroom.
- Unauthorized communication e.g. with another student.
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9. Once you have completed the assessment,

- Click on the "Exam Controls" button and choose "Submit Exam".
- Check off "I am ready to exit my exam" and click on "Exit" to upload your answers to the server.
- You will see a green confirmation window on your screen when the upload is successful. **Please keep this window on your screen.**
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- Please remain seated quietly in the meantime.

10. By clicking on the "Continue" button below, I acknowledge that I have read and fully understood this Exam Notice.

Thank You

Question 1

(MRQ) We would like to use polynomial regression to detect if an email is spam or not. The table below shows the training accuracy, validation accuracy and test accuracy when different polynomial orders are used. Which polynomial order should be selected?

order	Training accuracy	Validation accuracy	Test accuracy
1	0.90	0.90	0.98
2	0.94	0.98	0.95

3	0.96	0.95	0.92
4	0.98	0.93	0.90

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (b). Hyperparameter should be chosen according to validation performance.

Question 2

(MCQ) We would like to build a system that predicts if a user will like a movie before they watch it. We have a labeled dataset of past ratings, where each data sample is labeled as “like” or “dislike”. Which type of learning should be used?

- (a) Supervised Learning
- (b) Unsupervised Learning
- (c) Reinforcement Learning
- (d) None

Answer: (a). Labelled data should use supervised learning.

Question 3

(T/F) Given the following dataset for training:

Sample No.	$\mathbf{x}$	y
1	[1, 1.5]	1.0
2	[0.9, 2]	2.6
3	[3, 1]	3.0

where $\mathbf{x}$ is a 2-dimensional feature vector and y is the corresponding target value. If we fit a polynomial regression model of order 3, we can find a solution $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$ for the regression coefficients.

- (a) True
- (b) False

Answer: (b). $\mathbf{P}^T \mathbf{P}$ is not invertible.

Question 4

(MRQ) In electronics manufacturing, circuit defects such as broken traces, short circuits, or soldering issues can compromise performance and reliability. We would like to develop a binomial logistic regression model that can automatically detect if a circuit board image contains a defect. After training with a dataset of circuit board images labelled as either 0 meaning “no defect” or 1 meaning “defect”, a new image is present to the model and the output is 0.3. Which of the following statements is correct?

- (a) The model is 30% confident that the circuit board contains no defect.
- (b) The model predicts that the circuit board has a defect, because the output is not 0.
- (c) The model predicts that the circuit board has no defect, because the output is below 0.5.
- (d) None

Answer: (d).

(a) is wrong because the model is 30% confident that the circuit board contains defect.

(b) and (c) are wrong because no threshold is predefined, so we are not sure about the exact class prediction.

Question 5

(MRQ) Which of the following statements is/are correct?

- (a) K-means clustering is not suitable for datasets with spiral-shaped clusters.
- (b) Given the same set of data points, the number of well-separated clusters that can be formed is always no more than the number of center-based clusters that can be formed.
- (c) Running K-means clustering multiple times with different random initializations can guarantee reaching the global minimum solution.
- (d) Each data point can belong to multiple clusters for K-means clustering.
- (e) None

Answer: (a)(b)

(c) is wrong because it cannot guarantee reaching the global minimum solution. It only increases the likelihood.

Question 6

(T/F) A single-layer perceptron is unable to handle linearly inseparable data.

- (a) True
- (b) False

Answer: (a)

Question 7

(MRQ) Given the following confusion matrix for a binary classification model that differentiates between an image of dog and an image of cat, which of the following statements is/are correct? Assume the positive class is dog and the negative class is cat.

	$\hat{P}$ (predicted)	$\hat{N}$ (predicted)
P (actual)	45	20
N (actual)	5	30

- (a) There are 20 cat images that are wrongly classified. They are known as False Negatives.
- (b) The number of images of dogs is the same as the number of images of cats.
- (c) The classification accuracy of the model is 75%.
- (d) The precision of the model is 69.23%, rounded to two decimal places.
- (e) None

Answer: (c)

(a) is wrong because the number “20” in the confusion matrix means 20 dog images that are wrongly classified as cat.

(b) is wrong because the number of images of dogs is 75 while the number of images of cats is 35. They are not the same.

(d) is wrong because the precision is calculated by $TP/(TP+FP)=45/(45+5)=90\%$.

Question 8

(FITB) Given the following dataset for training:

Sample No.	$\mathbf{x}$	y
1	[3, 15, 50]	5.0
2	[5, 12, 68]	6.0
3	[2, 9, 60]	3.0
4	[4, 19, 80]	8.0

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target value. Fit a polynomial regression model of order 1. What is predicted output of a new data [7, 10, 70]? If the solution $\mathbf{w}^* = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$ does not exist, apply L2 regularization with the regularization parameter $\lambda = 0.1$.

Your Answer____(up to 4 decimal places)

Answer: 6.6603 ($P^T P$ is invertible)

Range: Min: 6.660 Max: 6.661

Please see “Quiz 3_A.ipynb”

Question 9

(FITB) Given the following dataset for training a logistic regression model for 3-class classification:

Sample No.	$\mathbf{x}$	y
1	[0.5, 1.2, -0.3]	1
2	[-1, 0.8, 1.5]	2
3	[2.3, -0.7, 0.5]	3
4	[0, 1.5, -1.0]	1

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target class label.

Assume the initial regression coefficient matrix $\mathbf{W} = \begin{bmatrix} w_{0,1} & w_{0,2} & w_{0,3} \\ w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix} =$

$\begin{bmatrix} 0 & 0 & 0 \\ 0.01 & -0.02 & 0.03 \\ 0.05 & 0.04 & -0.01 \\ -0.03 & 0.02 & 0.01 \end{bmatrix}$. Gradient descent is used for learning and learning rate is 0.1.

What is the value of the regression coefficient $w_{0,1}$ after the 1st iteration of gradient descent?

Your Answer: ____ (up to 4 decimal places)

Answer: 0.0163

Range: Min: 0.0162 Max: 0.0164

Please see “Quiz 3_A.ipynb”

Question 10

(FITB) Given the following dataset for training a multi-layer perceptron (MLP) of 2 layers for 3-class classification:

Sample No.	$\mathbf{x}$	y
1	[0.5, 1.2, -0.3]	1
2	[-1, 0.8, 1.5]	2
3	[2.3, -0.7, 0.5]	3
4	[0, 1.5, -1.0]	1

where $\mathbf{x}$ is a 3-dimensional feature vector and y is the corresponding target class label. The MLP has been constructed as $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \text{Softmax}([\mathbf{1}, \text{ReLU}(\mathbf{x}^T \mathbf{W}^{(1)})] \mathbf{W}^{(2)})$. Assume

the initial weight matrices $\mathbf{W}^{(1)} = \mathbf{W}^{(2)} = \begin{bmatrix} w_{0,1} & w_{0,2} & w_{0,3} \\ w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix} =$

$\begin{bmatrix} 0 & 0 & 0 \\ 0.01 & -0.02 & 0.03 \\ 0.05 & 0.04 & -0.01 \\ -0.03 & 0.02 & 0.01 \end{bmatrix}$. Gradient descent is used for learning and learning rate is 0.1.

What is the value of $w_{0,1}$ in $\mathbf{W}^{(1)}$ after the 1st iteration of gradient descent?

Your Answer: $\times 10^{-4}$ (up to 4 decimal places)

Answer: 1.6546

Range: Min: 1.65 Max: 1.66

Please see "Quiz 3_A.ipynb"

Question 11

(MCQ) A healthcare clinic collects data from 5 patients, including their age, height, and weight, and measures their blood pressure. They want to build a machine learning model to automatically identify patients who are at risk of high blood pressure. The data collected is shown on the table below. Which technique can be observed from this dataset that has been used during the data preparation?

Patient	Age	Height	Weight	Blood Pressure	Risk
1	25	163	50	120	No
2	30	175	0	135	Yes
3	0	168	65	140	Yes
4	22	180	72	110	No
5	28	174	68	125	No

- (a) Imputation
- (b) One-hot encoding
- (c) Normalization
- (d) Feature selection using Pearson's correction coefficient.

Answer: (a). For Patient 3, Age can never be 0. Similarly, for Patient 2, Weight can never be 0. The only possible explanation is that these two cells are used to be missing values. Then, values out of normal range (i.e., 0) are filled in these two cells. This is exactly the application of data imputation.

Question 12

(MRQ) Which of the following statements is/are correct. **Select all that apply.**

- (a) Least Squares Regression finds regression coefficients such that the predicted outputs are the orthogonal projection of the target output vector onto the column space of the training data matrix.
- (b) The solution obtained by Least Squares Regression is always a global minimum solution.
- (c) When the number of the samples n is smaller than the number of the input features d in the data matrix $\mathbf{X}_{n \times d}$, $\mathbf{X}^T \mathbf{X}$ is not invertible.
- (d) None

Answer: (a)(b)(c).

(c) is correct because $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}^T \mathbf{X})$. Since $n < d$, $\text{rank}(\mathbf{X}) \leq n < d$. Then, $\text{rank}(\mathbf{X}^T \mathbf{X}) = \text{rank}(\mathbf{X}) \leq n < d$. Since $\mathbf{X}^T \mathbf{X}$ is of size $d \times d$ and $\text{rank}(\mathbf{X}^T \mathbf{X}) < d$, $\mathbf{X}^T \mathbf{X}$ is not invertible.